

Part 6 Training LLaMA 3 on TPU - Counting Params and FLOPs

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i) $\text{FLOPs per step} = 6 \cdot \text{tokens} \cdot \text{params} = 6 \cdot 70B = 420B \text{ FLOPs/step}$

ii) Training for $15T$ tokens takes $15T \cdot 420B = 6.3e24 \text{ FLOPs}$

iii) $t_{\text{train}} = \frac{6.3e24}{4.59e4 \cdot 8960 \cdot 0.4} = 3.82e6 \text{ seconds}$

$\approx 44 \text{ days} \approx 6 \text{ weeks}$

iv) $B = 4M \text{ tokens}$, so if we store 4 chunks, (lets assume we store ~~assumed~~ post attention (BD), after up proj (2BF), after down

proj (BD) = $2(2(BP+BF))$ bytes per layer.

$$\rightarrow 2(2(4M(8192+28672))) \cdot 80$$

$\approx 47 \text{ TB}$

Params + opt state = $10 \cdot 70B = 700B \text{ bytes} \approx 700 \text{ GB}$

Total = 47.7 TB .

Each TPU has 96 GB . So, we need $\frac{47.7 \text{ TB}}{96 \text{ GB}} = 492 \text{ TPUs}$ at minimum.

* In the above way, they assume all checkpoints are of shape BD $\rightarrow$ This yields 225 TPUs or a little less than half.

v) Assuming we have 21.6TB total, then each chip stores $\frac{21.6TB}{8960} = \boxed{2.416TB}$

How to share LLaMA 3-70B for training

i) Assume we can't do context parallelism, $4M=13$, 1724 seg with of length 4096

This means we cannot split across sequence len

Since we can only split 1724 ways, each chip will hold $\approx 88x$ more GB, but this is still far less than 96GB. So we can train it (it will fit), but we would be using $1/8$ the ~~RAM~~ chips.

ii) As per the previous section, we'll say FSPP, unlike if we are comm bound.

We are comm bound if $\frac{4M}{8960} < \frac{2550}{3} \Leftrightarrow 446 \leq 840$, which is true.

So we are comm bound, and FSPP isn't very efficient.

iii) We will be compute bound as long as $\frac{4M}{8960} > \frac{2550^2}{2(28672)} \approx 113$,

so we can do mixed FSPP + TP

* Let $M_x=2$, $M_y=1$ as per prev section, then $X_{opt} = \sqrt{\frac{13}{F} \frac{M_x}{M_y}} N = \sqrt{\frac{4M \cdot 2 \cdot 8960}{28672 \cdot 1}}$

1120 FSPP, 8TP

≈ 1619 , so about 282x

~~2400 FSPP~~ (1120-9=8960)

Part 6 - Worker Problems

Q1) To scale to multiple pods, we need to use PCN.

Strategy: Use pure data parallelism across pods + FSDP/TP parallelism on ~~32x238~~ 1M B per DCT domain. This means that our per chip batch size is 112, which is less than 113, which means we are barely comm-bound with FSDP/TP locally (we are compute bound across pods).

i.e. $T_{\text{math}} \text{ for the pod} > T_{\text{comm}}$ for the data parallel comm between pods, but T_{comm} within pod $\geq T_{\text{math}}$ within pod. But t_{comm} is so close to t_{math} we may as well be compute bound / efficiently using hardware.

$$t_{\text{batch}} = \frac{6.3e24}{8960 * 4 * 0.4 * 4.59e14} \approx 11 \text{ days at } 40\% \text{ utilization}$$

Q2) $L = 126$ FFW ≈ 329.7

a) $D = 16,384$ Vocab ≈ 4 Billion

$F = 53,248$ Attn ≈ 71 Billion

$N = 128$

~~405~~ $404.7 \approx 405$

$K = 8$

$H = 128$

$V = 128,256$

▷ b) Let's assume $B = 4M$ tokens, the per node $B = 500K$.

▷ If we try pure FSPP, we have ~~$4 \times 1024 \times 1024 = 4194304$ per chip~~
 ~~8960~~

▷ ~~$468/8$~~ $468/8 = 59$ ^{cuts} per chip (with context parallelism).

▷ This is well below 840, so no pure FSPP.

▷ For FSPP/TP, we need $59 \times \frac{(2500)^2}{2F} = 61$. While this is not strictly true, it's close enough that we can expect good utilization.

▷ We also have that $500K > 80K$, so we are okay on the DCN front for remaining compute nodes.

▷ Putting it all together, we can expect to be mostly compute bound with Data Parallelism across nodes + FSPP/TP within nodes. This would take

$$\frac{6 * 1512 * 4059}{4.59e4 * 8960 * 8 * 0.4} = 2.8e6 \text{ sec} \approx 32 \text{ days at } 40\% \text{ util}$$

~~Scalability doesn't necessarily scale with larger~~